

Prenatal Tobacco Exposure and Neonatal Birth Weight

A Frequentist Comparative and Regression Analysis of the Baystate Low Birth Weight Study

Item	Detail
Research question	Is maternal smoking during pregnancy associated with lower infant birth weight, and does the association persist after covariate adjustment?
Data	birthwt (MASS, R): 189 singleton births, Baystate Medical Center, 1986
Methods	Welch t-test, Mann-Whitney U, Cohen's d / Hedges' g, OLS regression (crude and adjusted), residual diagnostics
Software	R (MASS, ggplot2)
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Abstract

Background. Low birth weight is a major predictor of neonatal morbidity and mortality, and maternal smoking is among the most widely studied modifiable risk factors. **Objective.** To estimate the association between maternal smoking during pregnancy and infant birth weight, and to assess whether the association is robust to the choice of test and to adjustment for maternal age, maternal weight and race. **Methods.** We analysed 189 births (115 non-smokers, 74 smokers) from the Baystate Low Birth Weight Study. Normality was assessed with Shapiro-Wilk tests and QQ plots. Mean birth weight was compared using Welch's two-sample t-test, with the Mann-Whitney U test as a rank-based robustness check. Effect size was summarised by Cohen's d. Crude and covariate-adjusted ordinary least squares (OLS) models were fitted, and the adjusted model was examined with residual, leverage and multicollinearity diagnostics.

Results. Mean birth weight was 3055.7 g (SD 752.7) among non-smokers and 2771.9 g (SD 659.6) among smokers, a raw difference of 283.8 g (Welch $t = 2.73$, $df = 170.1$, $p = 0.007$; 95% CI 78.6 to 489.0 g). The Mann-Whitney test agreed ($W = 5249.5$, $p = 0.007$), and Cohen's d was 0.40. In the crude regression the smoking coefficient was -283.8 g (95% CI -494.8 to -72.8). After adjustment for age, maternal weight and race it increased in magnitude to -401.7 g (95% CI -617.3 to -186.2 , $p < 0.001$), a 41.6% change that is consistent with negative confounding by race. **Conclusion.** Maternal smoking was associated with substantially lower birth weight, and the finding was robust across parametric, non-parametric and adjusted analyses. Because the data are observational and from a single hospital, the result describes association rather than causation.

Key findings at a glance

• Raw gap: **283.8 g** • Adjusted gap: **401.7 g** • Cohen's d : **0.40** • Welch $p = 0.007$ • Mann-Whitney $p = 0.007$ • Adjusted R^2 : **0.125**

1. Introduction and Objectives

Birth weight under 2500 g is classified as low birth weight and is strongly associated with infant mortality, impaired development and later chronic disease. Maternal tobacco use during pregnancy is a well-established correlate of reduced fetal growth. The Baystate Low Birth Weight Study, a classic teaching dataset in applied statistics, provides a compact sample for quantifying this association with standard inferential tools.

This report has three objectives: (i) to test whether mean birth weight differs between babies of smoking and non-smoking mothers; (ii) to verify that the conclusion does not depend on the distributional assumptions of the t-test; and (iii) to determine whether the association persists after accounting for other maternal characteristics. Let μ_1 and μ_2 denote the population mean birth weights of non-smokers and smokers. The primary hypotheses are

$$H_0 : \mu_1 = \mu_2 \quad \text{versus} \quad H_1 : \mu_1 \neq \mu_2 \quad (1)$$

tested at a two-sided significance level of $\alpha = 0.05$. All confidence intervals are 95% intervals.

2. Data and Study Design

The dataset `birthwt` is distributed with the MASS package. It records 189 births at Baystate Medical Center, Springfield, Massachusetts, in 1986, with ten variables and no missing values. The outcome is birth weight in grams (`bwt`) and the exposure is a binary indicator of maternal smoking during pregnancy (`smoke`). Race was recoded as a three-level factor with White as the reference category.

Variable	Type	Description
bwt	Continuous (g)	Birth weight (outcome)
smoke	Binary	Maternal smoking during pregnancy (exposure): 0 = non-smoker, 1 = smoker
age	Continuous (years)	Maternal age (covariate)
lwt	Continuous (lb)	Maternal weight at last menstrual period (covariate)
race	Categorical (3)	White, Black, Other (covariate; White is reference)
low, ptl, ht, ui, ftv	Various	Low-birth-weight flag, previous premature labours, hypertension, uterine irritability, first-trimester visits (not modelled)

Table 1. Variables used in the analysis.

The sample contained 115 non-smokers (60.8%) and 74 smokers (39.2%). Because the study is observational, smoking status was not randomised, so differences in other maternal characteristics between the groups can confound a crude comparison. Section 7 returns to this point.

3. Statistical Methods

3.1 Normality assessment

Within each smoking group, normality of birth weight was assessed graphically by normal QQ plots and formally by the Shapiro-Wilk test, whose null hypothesis is that the sample is drawn from a normal population. Given the moderate sample sizes, the Central Limit Theorem additionally supports inference on the sample mean.

3.2 Welch two-sample t-test

Let $\bar{X}_1$, s_1^2 , n_1 and $\bar{X}_2$, s_2^2 , n_2 be the sample means, variances and sizes. Welch's statistic does not assume equal variances:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}} \quad (2)$$

with degrees of freedom approximated by the Welch-Satterthwaite formula

$$\nu = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)} \quad (3)$$

Because Welch's test remains valid whether or not the variances are equal, no preliminary variance test was required. A $(1 - \alpha)$ confidence interval for the mean difference is $(\bar{X}_1 - \bar{X}_2) \pm t_{1-\alpha/2, \nu} \times \text{SE}$.

3.3 Mann-Whitney U test

As a distribution-free check, the Mann-Whitney (Wilcoxon rank-sum) test pools both samples, ranks the observations, and compares rank sums. With R_1 the sum of ranks in group 1,

$$U = R_1 - \frac{n_1(n_1 + 1)}{2} \quad (4)$$

The null hypothesis is that a randomly chosen observation from one group is equally likely to exceed or fall below one from the other. The ratio $U/(n_1n_2)$ estimates the common-language effect size, the probability that a randomly chosen non-smoker's baby is heavier than a randomly chosen smoker's baby. R also reports the Hodges–Lehmann location shift, the median of all pairwise differences, with a confidence interval.

3.4 Effect size

Cohen's d expresses the mean difference in pooled standard deviation units, and Hedges' g applies a small-sample bias correction J :

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \quad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (5)$$

$$g = Jd, \quad J = 1 - \frac{3}{4(n_1 + n_2) - 9} \quad (6)$$

Conventional benchmarks are 0.2 (small), 0.5 (medium) and 0.8 (large).

3.5 Linear regression

Two OLS models were fitted. Model 1 is the crude model, and Model 2 adjusts for maternal age, maternal weight and race:

$$BWT_i = \beta_0 + \beta_1 \text{SMOKE}_i + \varepsilon_i \quad (7)$$

$$BWT_i = \beta_0 + \beta_1 \text{SMOKE}_i + \beta_2 \text{AGE}_i + \beta_3 \text{LWT}_i + \beta_4 \text{BLACK}_i + \beta_5 \text{OTHER}_i + \varepsilon_i \quad (8)$$

with errors assumed independent and $\varepsilon_i \sim N(0, \sigma^2)$. In matrix form $y = X\beta + \varepsilon$, and the OLS estimator and its covariance matrix are

$$\hat{\beta} = (X^T X)^{-1} X^T y, \quad \text{Var}(\hat{\beta}) = \sigma^2 (X^T X)^{-1} \quad (9)$$

Overall fit is summarised by $R^2 = 1 - \text{SSE}/\text{SST}$, the adjusted $R^2 = 1 - (1 - R^2)(n - 1)/(n - p - 1)$, and the overall F statistic $F = (R^2/p) / ((1 - R^2)/(n - p - 1))$ on p and $n - p - 1$ degrees of freedom. In Model 1 β_1 equals the difference in group means, so the regression and the t-test address the same contrast.

3.6 Confounding and diagnostics

Confounding was assessed by the relative change in the exposure coefficient between the crude and adjusted models,

$$\Delta\% = 100 \times \frac{\hat{\beta}_{1, \text{adj}} - \hat{\beta}_{1, \text{crude}}}{\hat{\beta}_{1, \text{crude}}} \quad (10)$$

where a change above about 10% is conventionally taken to indicate meaningful confounding. Model 2 was examined using residuals versus fitted values, a normal QQ plot of standardised residuals, a scale-location plot, leverage and Cook's distance, the Breusch–Pagan test for heteroscedasticity, and variance inflation factors $\text{VIF}_j = 1/(1 - R_j^2)$ for multicollinearity.

4. Descriptive Results

Statistic	Non-smoker	Smoker	Difference	Note
n	115	74	—	60.8% / 39.2%
Mean birth weight (g)	3055.7	2771.9	283.8	Non-smoker higher
SD (g)	752.7	659.6	—	Similar spread
Median (g)	3100.0	2775.5	324.5	
IQR (g)	1112.5	875.3	—	
Range (g)	1021 – 4990	709 – 4238	—	
Birth weight < 2500 g	25.2%	40.5%	15.3 pts	Raw proportions

Table 2. Birth weight by maternal smoking status.

Babies of smoking mothers were lighter on average by 283.8 g, and the median difference of 324.5 g points the same way. Group spreads were comparable (SD 752.7 g versus 659.6 g). The proportion below the 2500 g low-birth-weight threshold was 25.2% among non-smokers and 40.5% among smokers. Figure 1 shows the full distributions, and Figure 2 compares each histogram with a fitted normal density.

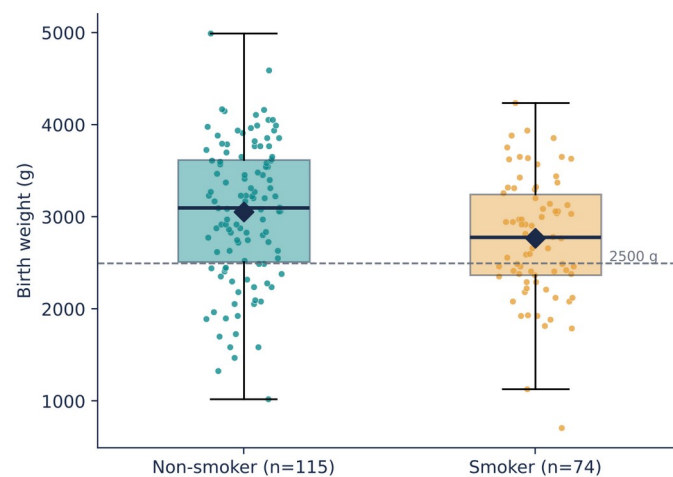


Figure 1. Birth weight by smoking status. Diamonds mark group means; dashed line marks 2500 g.

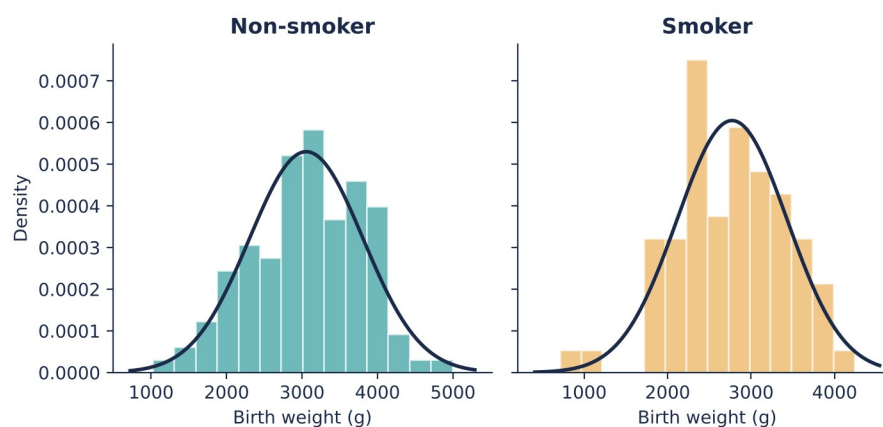


Figure 2. Histograms with fitted normal densities.

5. Assumption Checks

The histograms in Figure 2 are unimodal and approximately symmetric, with sample skewness of about -0.29 in both groups and excess kurtosis between -0.27 and 0.32 , all close to the normal values of zero. The QQ plots in Figure 3 follow the reference line closely, with only minor departures in the extreme tails.

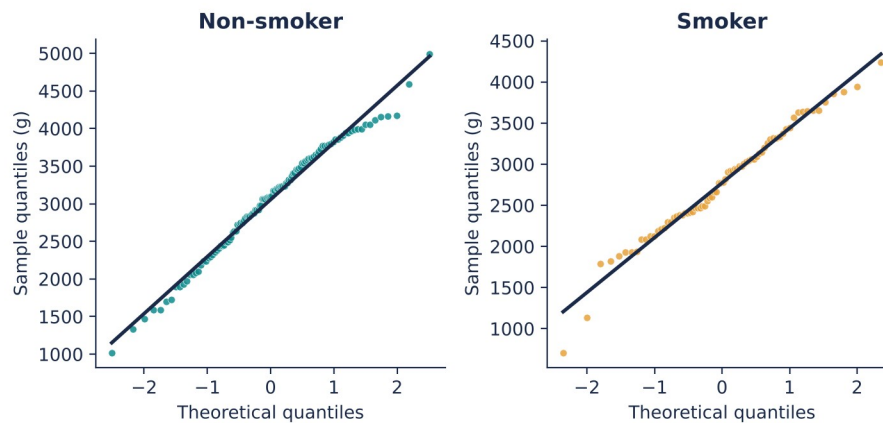


Figure 3. Normal QQ plots by smoking group.

Test	Statistic	p-value	Decision
Shapiro-Wilk, non-smokers	$W = 0.987$	0.334	No evidence against normality
Shapiro-Wilk, smokers	$W = 0.983$	0.419	No evidence against normality
Levene (equal variances)	$F = 1.456$	0.229	No evidence of unequal variance

Table 3. Distributional assumption tests.

Neither Shapiro-Wilk test rejects normality, and Levene's test shows no evidence of unequal variances. The assumptions behind the t-test therefore hold approximately. Because Welch's version was used regardless, the conclusions do not depend on the equal-variance result, and the Levene result is reported only as supporting information.

6. Hypothesis Tests and Effect Size

Analysis	Result	Interpretation
Welch t-test	$t = 2.730$, $df = 170.1$, $p = 0.0070$	Reject H_0 at 5%
Mean difference (95% CI)	283.8 g (78.6 to 489.0)	CI excludes 0
Standard error of difference	104.0 g	—
Mann-Whitney U	$W = 5249.5$, $p = 0.0068$	Agrees with t-test
Hodges-Lehmann shift (95% CI)	306.2 g (85.0 to 512.0)	Robust location shift
Common-language effect size	$U/(n_1 n_2) = 0.617$	61.7% chance non-smoker's baby is heavier
Cohen's d (95% CI)	0.395 (0.10 to 0.69)	Small-to-moderate
Hedges' g	0.394	Bias-corrected, nearly identical

Table 4. Two-group comparison of mean birth weight.

Both tests reject the null hypothesis of no difference at the 5% level, and they do so with almost identical p-values (0.0070 and 0.0068). The parametric and rank-based estimates of the gap (283.8 g and 306.2 g) are of similar size. Agreement between the two approaches shows that the conclusion does not rest on the normality assumption.

The effect size is small-to-moderate: Cohen's *d* of 0.395 lies between the conventional small (0.2) and medium (0.5) benchmarks, and its confidence interval of 0.10 to 0.69 is wide, reflecting the modest sample. In probabilistic terms, a randomly chosen baby of a non-smoking mother outweighs a randomly chosen baby of a smoking mother about 62% of the time. The 95% confidence interval for the mean difference is wide (79 to 489 g), so while the direction of the effect is clear, its magnitude is estimated imprecisely.

7. Regression Analysis

7.1 Crude model

Model 1 regresses birth weight on smoking status alone. The intercept of 3055.7 g reproduces the non-smoker mean, and the smoking coefficient of -283.8 g (SE 107.0, *t* = -2.65, *p* = 0.0087) reproduces the difference found in the *t*-test. Smoking alone explains 3.6% of the variance in birth weight ($R^2 = 0.036$, $F(1, 187) = 7.04$, *p* = 0.0087), which is expected given the many other determinants of fetal growth.

7.2 Adjusted model

Term	Estimate (g)	SE	<i>t</i>	<i>p</i>	95% CI
Intercept	2839.4	321.4	8.83	< 0.001	2205.2 to 3473.6
Smoker	-401.7	109.2	-3.68	< 0.001	-617.3 to -186.2
Age (per year)	-1.9	9.8	-0.20	0.843	-21.3 to 17.4
Maternal weight (per lb)	4.0	1.7	2.30	0.022	0.6 to 7.4
Black vs White	-510.5	157.1	-3.25	0.001	-820.4 to -200.6
Other vs White	-398.6	119.6	-3.33	0.001	-634.6 to -162.7

Table 5. Adjusted OLS model (Model 2). Residual SE = 682.1 on 183 df; $R^2 = 0.148$; adjusted $R^2 = 0.125$; $F(5, 183) = 6.37$, *p* = 1.8×10^{-5} .

After adjustment, smoking remained strongly associated with lower birth weight: babies of smokers were on average 401.7 g lighter than babies of non-smokers of the same age, weight and race (95% CI 186.2 to 617.3 g lighter). Maternal age was not associated with birth weight (*p* = 0.843). Each additional pound of maternal weight was associated with about 4 g higher birth weight (*p* = 0.022). Compared with White mothers, Black and Other mothers had babies who were on average 510.5 g and 398.6 g lighter, respectively. The model explains 14.8% of the variance (adjusted $R^2 = 0.125$), and the overall *F* test is highly significant.

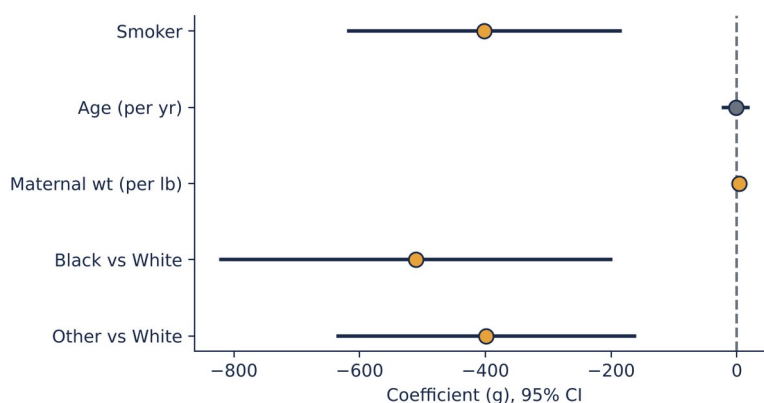


Figure 4. Adjusted coefficients with 95% confidence intervals. Amber points are statistically significant.

7.3 Confounding by race

Applying equation (10), the smoking coefficient changed by $(-401.7 - (-283.8)) / (-283.8) = +41.6\%$, well beyond the 10% rule of thumb. The adjusted effect is larger, not smaller, which is the signature of negative confounding. The cross-tabulation explains why. Race is strongly associated with smoking in this sample ($\chi^2 = 21.78$ on 2 df, $p < 0.001$): 54.2% of White mothers smoked, compared with 38.5% of Black mothers and 17.9% of Other mothers. White mothers also had the heaviest babies on average (3102.7 g, versus 2719.7 g and 2805.3 g). Smoking was therefore concentrated in the group with the highest baseline birth weights, which masked part of its effect in the crude comparison. Adjusting for race removes this masking.

Race	n	Smokers (%)	Mean bwt (g)	Role
White	96	54.2	3102.7	Reference group
Black	26	38.5	2719.7	
Other	67	17.9	2805.3	

Table 6. Smoking prevalence and mean birth weight by race.

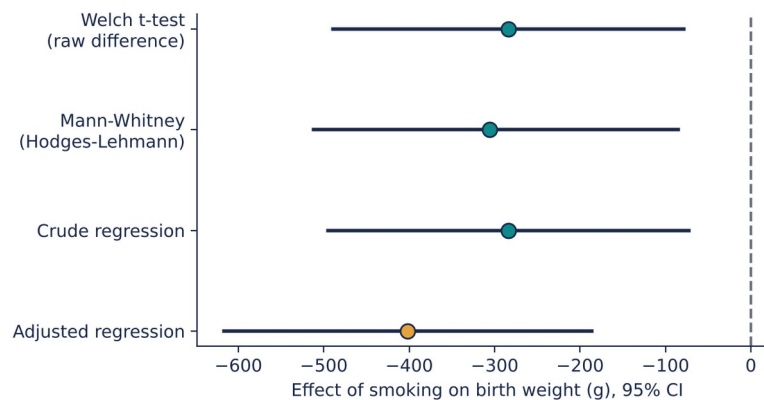


Figure 5. Estimated smoking effect across methods, with 95% confidence intervals. The adjusted estimate is the largest.

8. Model Diagnostics

Figure 6 shows the standard diagnostic plots for Model 2. Residuals scatter evenly around zero with no clear curvature, supporting the linearity assumption. The QQ plot of standardised residuals tracks the reference line, and the Shapiro-Wilk test on the residuals is not significant ($p = 0.266$). The scale-location plot shows no strong trend, and the Breusch-Pagan test gives no evidence of heteroscedasticity ($LM = 7.19$, $p = 0.207$).

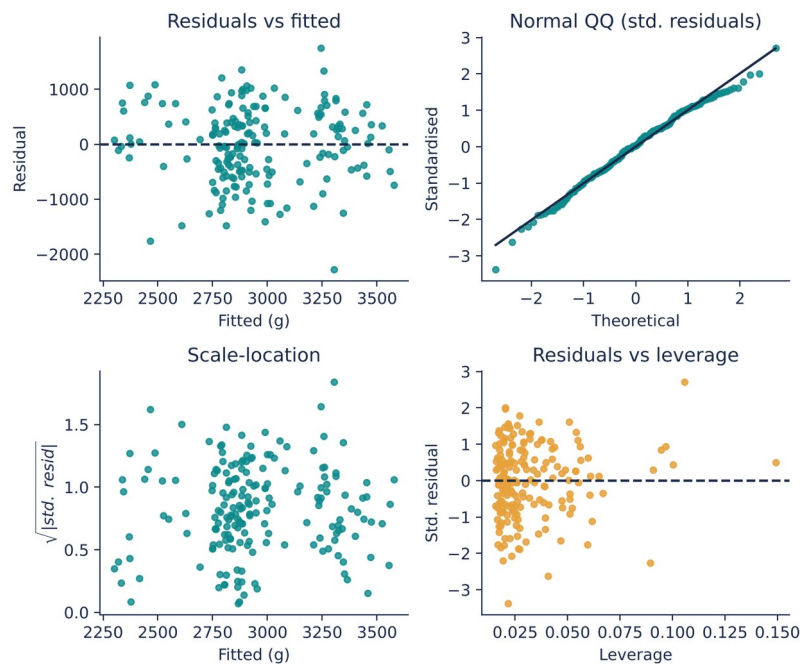


Figure 6. Diagnostic plots for the adjusted regression model.

Diagnostic	Value	Assessment
Breusch-Pagan (heteroscedasticity)	p = 0.207	Constant variance plausible
Shapiro-Wilk on residuals	p = 0.266	Residuals roughly normal
Variance inflation factors	Max 1.33 (all < 1.4)	No multicollinearity concern
Maximum Cook's distance	0.144	Well below 1; no dominant point
Maximum leverage	0.149	Moderate; no extreme leverage point
Standardised residuals above 2 in size	7 of 189 (3.7%)	Close to the roughly 5% expected

Table 7. Summary of regression diagnostics for Model 2.

Variance inflation factors are all below 1.4, so the covariates are not collinear enough to destabilise the estimates. The largest Cook's distance is 0.144, far below the conventional threshold of 1, so no single observation drives the fitted model. Together these checks indicate that the OLS assumptions are reasonably satisfied and that the reported inference is trustworthy.

9. Discussion

Three independent analytic routes lead to the same conclusion. The Welch t-test and the Mann-Whitney test both detect a difference of roughly 280 to 310 g, the regression coefficient reproduces the t-test difference exactly, and the adjusted model strengthens rather than weakens the association. The consistency of the evidence, together with the satisfied diagnostics, makes it unlikely that the finding is an artefact of a single modelling choice.

The most instructive result is the shift from -283.8 g to -401.7 g after adjustment. A naive reading of the crude analysis would understate the smoking association by roughly 118 g. This illustrates why crude two-group comparisons in observational data should be accompanied by adjusted analyses, and why the direction of confounding cannot be assumed in advance.

Limitations

- **Association, not causation.** Smoking was not randomised, and unmeasured confounders (nutrition, socioeconomic status, alcohol use) may remain.
- **Exposure measurement.** Smoking is binary, so dose-response, timing and intensity cannot be examined, and self-reported smoking may be misclassified.
- **Limited sample and setting.** There are 189 births from a single hospital in 1986, so confidence intervals are wide and generalisation to other populations or periods is uncertain.
- **Incomplete adjustment.** Hypertension, previous premature labour and uterine irritability are available but were not included; the adjusted model explains only about 15% of the variance in birth weight.
- **Race as a covariate.** Race here is a coarse proxy for social and environmental factors rather than a biological explanation.

Possible extensions

Natural next steps include adding the remaining clinical covariates, testing a smoking $\times$ race interaction, modelling the binary low-birth-weight indicator with logistic regression, and using bootstrap confidence intervals as a further robustness check.

10. Conclusion

Summary of findings

Babies of mothers who smoked during pregnancy weighed about **284 g less** on average than babies of non-smokers (Welch $p = 0.007$; 95% CI 79 to 489 g). The Mann-Whitney test agreed ($p = 0.007$), and the effect size was small-to-moderate (Cohen's $d = 0.40$). After adjusting for maternal age, weight and race, the association was larger at **402 g** (95% CI 186 to 617 g; $p < 0.001$), consistent with negative confounding by race. Regression diagnostics gave no cause for concern. The results describe an association in a single-hospital observational sample and do not establish causation.

Reproducibility

All results were produced in R using the MASS and ggplot2 packages. The data load directly from MASS, so the script runs with no external files. Additional diagnostics (Levene, Breusch–Pagan, VIF, Cook's distance) can be reproduced with the car and lmtest packages, for example `car::leveneTest()`, `lmtest::bptest()`, `car::vif()` and `cooks.distance()`.

References

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